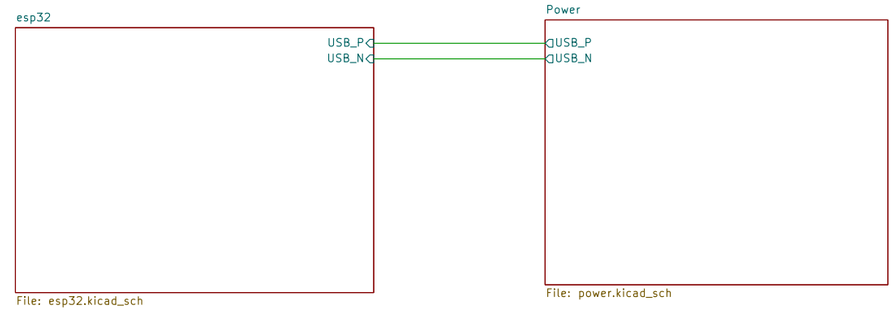


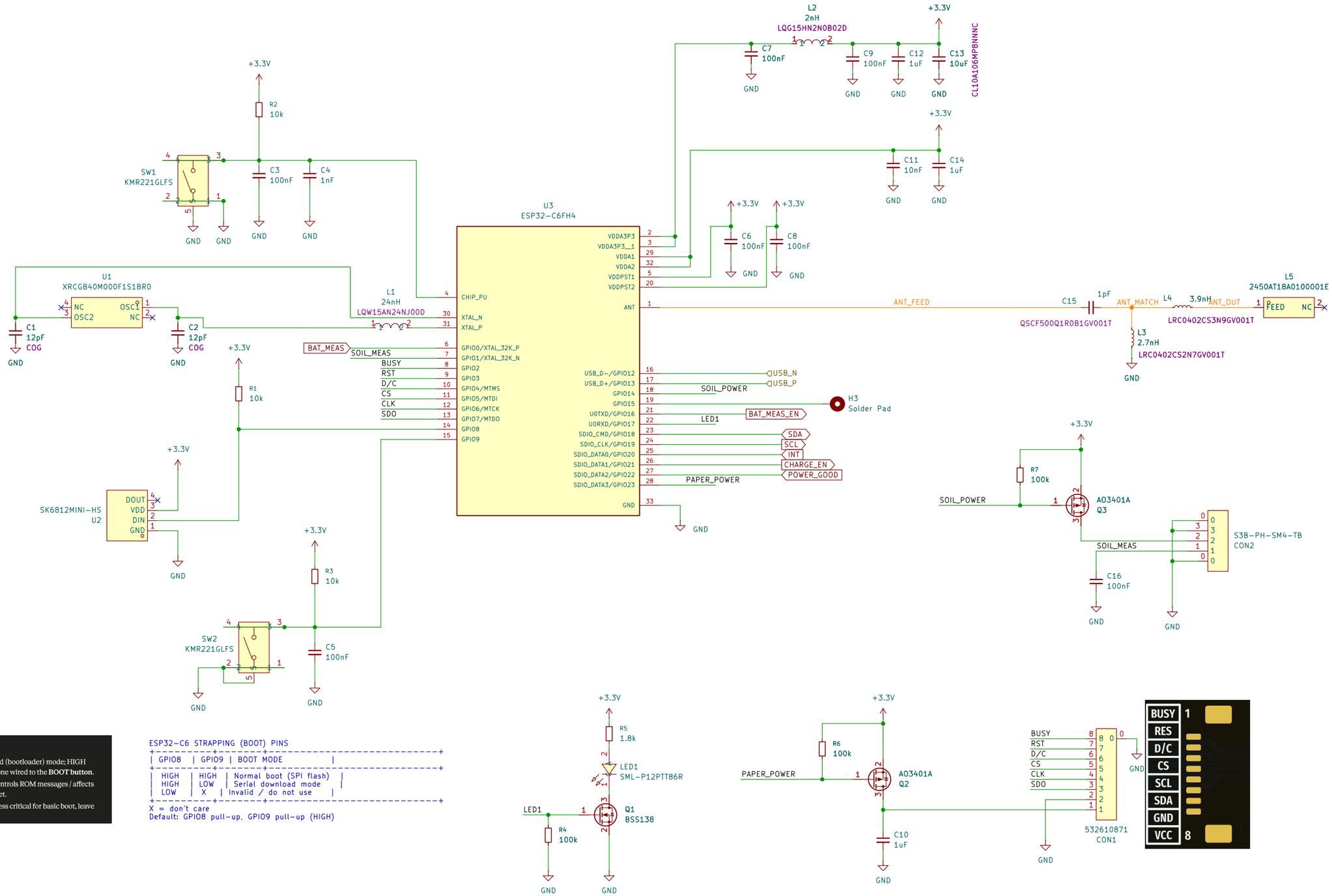


Author: Robin Bühler

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File: overview.kicad_sch

Title: Soil Sensor

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For the ESP32-C6, the strapping (boot) GPIOs are:

- **GPIO9 -** Boot mode select. Held LOW at reset → download (bootloader) mode; HIGH (default, internal pull-up) → normal SPI boot. This is the one wired to the **BOOT button.**
- **GPIO8 -** Must be HIGH at reset for normal boot. It also controls ROM messages / affects boot; it should have a pull-up and not be pulled low at reset.
- **GPIO15 -** JTAG/USB signal source select strapping pin (less critical for basic boot, leave default).

ESP32-C6 STRAPPING (BOOT) PINS

GPIO8	GPIO9	BOOT MODE
HIGH	HIGH	Normal boot (SPI flash)
HIGH	LOW	Serial download mode
LOW	X	Invalid / do not use

X = don't care
Default: GPIO8 pull-up, GPIO9 pull-up (HIGH)

Genuinely free and unencumbered on this symbol:
GPIO0, 1, 2, 3, 4, 5, 6, 7, 10, 11, 14, and the GPIO16-23 block (U0TXD/RXD, SDIO_CMD/CLK, SDIO_DATA0-3). A few sub-notes:

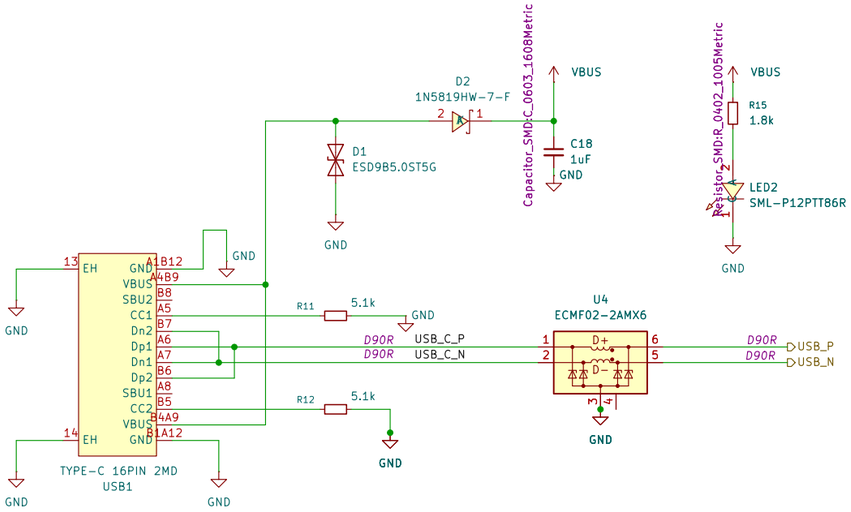
GPIO16/GPIO17 are the default UART0 TX/RX. Fine to use as GPIO, but if you want serial console/logging you'll want to leave them for that.
GPIO0/GPIO1 double as the 32 kHz crystal pins (XTAL_32K) — only relevant if you plan to add an external 32.768 kHz crystal for RTC accuracy. Otherwise free.
GPIO4/5/6/7 carry JTAG (MTMS/MTDI/MTCK/MTDO) and ADC functions — all fine as regular GPIO if you don't need on-chip JTAG debug.

Author: Robin Bühler

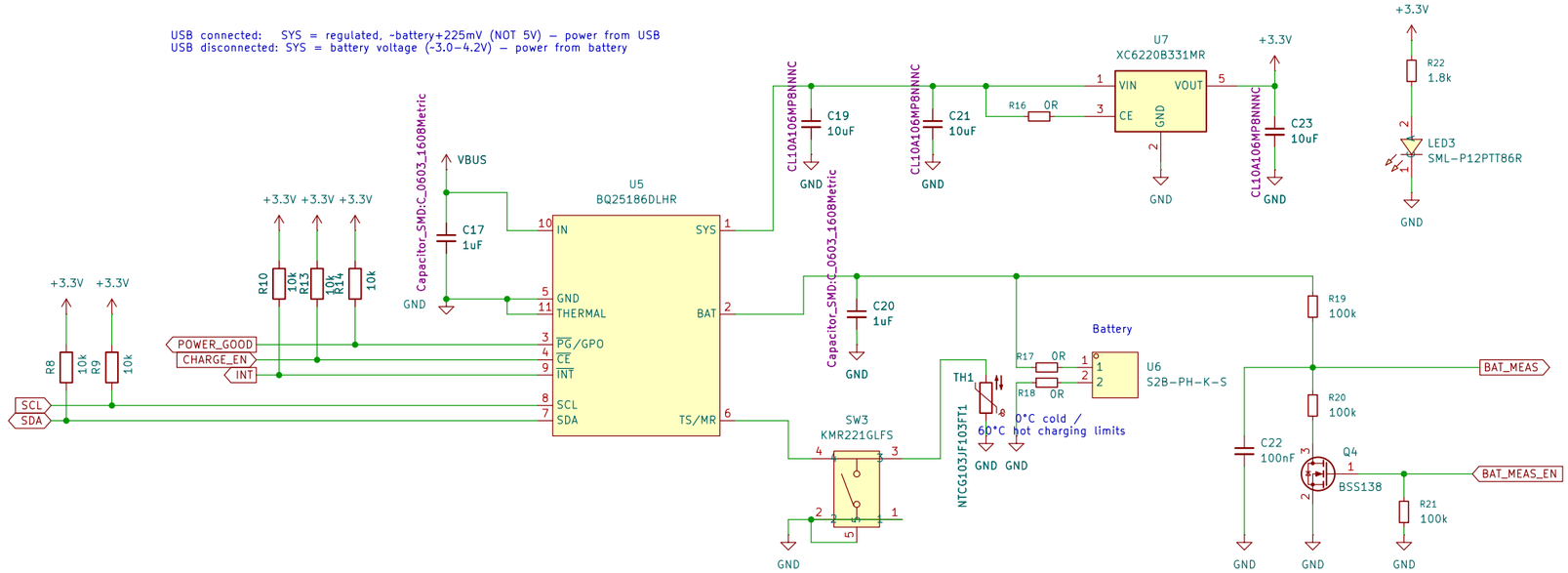
Sheet: /esp32/
File: esp32.kicad_sch

Title: Soil Sensor

Size: A3	Date: 2026-06-26	**Rev: 0.1.0**
KiCad E.D.A. 10.0.1		Id: 2/3



USB connected: SYS = regulated, ~battery+225mV (NOT 5V) — power from USB
USB disconnected: SYS = battery voltage (~3.0–4.2V) — power from battery



Author: Robin Bühler	
Sheet: /Power/	
File: power.kicad_sch	
Title: Soil Sensor	
Size: A3	Date: 2026-06-26
KiCad E.D.A. 10.0.1	Id: 3/3
	Rev: 0.1.0